

LINEAR MODELS FOR GROUPED DATA

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Summary: The analysis of a grouped response variable with an underlying continuous distribution is considered. A cross classification of explanatory variable(s) results into a single-factor, two-factor or even a multifactor design with T cells containing a frequency distribution with k class intervals. Employing a maximum likelihood estimation procedure, distribution fitting is done simultaneously over the T cells. Coherent to the distribution fitting a multifactor model is formulated to evaluate the effect of the explanatory variables on the grouped response variable. Standard errors are obtained from the multivariate delta theorem.

1. Introduction

Grouped data is a common phenomenon in many disciplines where typical continuous variables such as age or income are categorised into class intervals. The implication is that the usual statistical techniques for continuous response variables can no longer be applied. Researchers are often tempted to ignore the underlying continuous distribution of the grouped response variable which results in valuable information being ignored.

The analysis of multifactor designs in the case where the response variable is only observed in grouped format is addressed in this paper. Independent

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predictors may be included by way of a single-factor, two-factor or even a multifactor model. Specific underlying distributions for the response variable are considered to include the normal and log-logistic distributions. The basic statistical procedure described in Crowther, Crafford and Matthews (2007) for fitting a single distribution to grouped data, is extended to multifactor designs. Maximum likelihood estimates of parameters and effects in such models are determined in a simple and elegant way.

2. Formulation

Consider a grouped response frequency vector observed for each of T different cells in a multi-factor design. Each cell contains the frequencies of k class intervals. The basic formulation of the grouped response variable, to be modeled over the T cells of the multifactor design is summarized in Table 1.

Table 1 Grouped data in a multifactor design

Cells	Class interval				
	$(-\infty, x_1)$	$[x_1, x_2)$	$\cdots$	$[x_{k-2}, x_{k-1})$	$[x_{k-1}, \infty)$
1	f_{11}	f_{12}	$\cdots$	$f_{1,k-1}$	f_{1k}
2	f_{21}	f_{22}	$\cdots$	$f_{2,k-1}$	f_{2k}
$\vdots$	$\vdots$	$\vdots$	$\cdots$	$\vdots$	$\vdots$
T	f_{T1}	f_{T2}	$\cdots$	$f_{T,k-1}$	f_{Tk}

Let

$$\mathbf{F} = \begin{pmatrix} f_{11} & f_{12} & \cdots & f_{1,k-1} \\ f_{21} & f_{22} & \cdots & f_{2,k-1} \\ \vdots & \vdots & \cdots & \vdots \\ f_{T1} & f_{T2} & \cdots & f_{T,k-1} \end{pmatrix} = \begin{pmatrix} \mathbf{f}'_1 \\ \mathbf{f}'_2 \\ \vdots \\ \mathbf{f}'_T \end{pmatrix} : T \times (k-1) \quad (1)$$

be the matrix of frequencies where the rows of $\mathbf{F}$ denote the T cells of the multifactor design and the columns of $\mathbf{F}$ denote the first $(k-1)$ class intervals

with upper class limits

$$\mathbf{x}' = (x_1 \quad x_2 \quad \cdots \quad x_{k-1}) . \quad (2)$$

Define

$$\mathbf{f} = \text{vec}(\mathbf{F}) = \begin{pmatrix} \mathbf{f}_1 \\ \mathbf{f}_2 \\ \vdots \\ \mathbf{f}_T \end{pmatrix} : T(k-1) \times 1 \quad (3)$$

as the so-called concatenated frequency vector where the T rows of $\mathbf{F}$ in (1) are stacked row by row in a single column vector. It is assumed that the vector $\mathbf{f}$ is a product multinomial vector with fixed subtotals

$$\mathbf{n}' = (n_1 \quad n_2 \quad \cdots \quad n_T) \quad (4)$$

allocated to each of the T cells. Dividing the frequencies in each cell by its corresponding sample size in (4), leads to the concatenated vector of relative frequencies

$$\mathbf{p}_0 = \begin{pmatrix} \mathbf{p}_{01} \\ \mathbf{p}_{02} \\ \vdots \\ \mathbf{p}_{0T} \end{pmatrix} = \begin{pmatrix} \frac{1}{n_1} \mathbf{f}_1 \\ \frac{1}{n_2} \mathbf{f}_2 \\ \vdots \\ \frac{1}{n_T} \mathbf{f}_T \end{pmatrix} = ((\text{diag}(\mathbf{n}))^{-1} \otimes \mathbf{I}_{k-1}) \cdot \mathbf{f} \quad (5)$$

with vector of expected values and covariance matrix

$$E(\mathbf{p}_0) = \begin{pmatrix} \pi_{01} \\ \pi_{02} \\ \vdots \\ \pi_{0T} \end{pmatrix} = \boldsymbol{\pi}_0 \quad \text{Cov}(\mathbf{p}_0) = \begin{pmatrix} \mathbf{V}_{01} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{V}_{02} & \mathbf{0} & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{V}_{0T} \end{pmatrix} = \mathbf{V}_0$$

where the covariance matrix for the t -th cell under a multinomial sampling scheme is

$$\text{Cov}(\mathbf{p}_{0t}) = \frac{1}{n_t} (\text{diag}(\boldsymbol{\pi}_{0t}) - \boldsymbol{\pi}_{0t} \boldsymbol{\pi}_{0t}') = \mathbf{V}_{0t} \quad (6)$$

for $t = 1, 2, \dots, T$.

The concatenated vector of cumulative relative frequencies follows

$$\mathbf{p} = \begin{pmatrix} \mathbf{p}_1 \\ \mathbf{p}_2 \\ \vdots \\ \mathbf{p}_T \end{pmatrix} = \begin{pmatrix} \mathbf{C}\mathbf{p}_{01} \\ \mathbf{C}\mathbf{p}_{02} \\ \vdots \\ \mathbf{C}\mathbf{p}_{0T} \end{pmatrix} = (\mathbf{I}_T \otimes \mathbf{C}) \mathbf{p}_0, \quad \mathbf{C} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 1 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{pmatrix} : (k-1) \times (k-1). \quad (7)$$

The vector of expected values and covariance matrix, for the vector $\mathbf{p}$ are

$$E(\mathbf{p}) = (\mathbf{I}_T \otimes \mathbf{C}) \boldsymbol{\pi}_0 = \begin{pmatrix} \pi_1 \\ \pi_2 \\ \vdots \\ \pi_T \end{pmatrix} = \boldsymbol{\pi} \quad (8)$$

and

$$\text{Cov}(\mathbf{p}) = (\mathbf{I}_T \otimes \mathbf{C}) \mathbf{V}_0 (\mathbf{I}_T \otimes \mathbf{C})' = \begin{pmatrix} \mathbf{V}_1 & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{V}_2 & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{V}_T \end{pmatrix} = \mathbf{V} \quad (9)$$

respectively, with $\mathbf{V}_t = \mathbf{C}\mathbf{V}_{t0}\mathbf{C}'$ for $t = 1, 2, \dots, T$.

3. Estimation

3.1 ML estimation procedure

The random vector $\mathbf{p}$ in (7) belongs to the exponential family. Let $\mathbf{g}(\boldsymbol{\pi})$ be a continuous vector valued function of $\boldsymbol{\pi}$, for which the first order partial derivatives, $\mathbf{G}_\pi = \frac{\partial \mathbf{g}(\boldsymbol{\pi})}{\partial \boldsymbol{\pi}}$ with respect to $\boldsymbol{\pi}$ exist. The ML estimate of $\boldsymbol{\pi}$, subject to the constraints $\mathbf{g}(\boldsymbol{\pi}) = \mathbf{0}$ is obtained iteratively from

$$\hat{\boldsymbol{\pi}} = \mathbf{p} - (\mathbf{G}_\pi \mathbf{V})' (\mathbf{G}_p \mathbf{V} \mathbf{G}_\pi')^* \mathbf{g}(\mathbf{p}) \quad (10)$$

where $\mathbf{G}_p = \left. \frac{\partial \mathbf{g}(\boldsymbol{\pi})}{\partial \boldsymbol{\pi}} \right|_{\boldsymbol{\pi}=\mathbf{p}}$ and $(\mathbf{G}_p \mathbf{V} \mathbf{G}_\pi')^*$ is a generalized inverse of $(\mathbf{G}_p \mathbf{V} \mathbf{G}_\pi')$. Further, the asymptotic covariance matrix of $\hat{\boldsymbol{\pi}}$, under $\mathbf{g}(\boldsymbol{\pi}) = \mathbf{0}$,

is

$$\text{Cov}(\hat{\pi}) \cong \mathbf{V} - (\mathbf{G}_\pi \mathbf{V})' (\mathbf{G}_\pi \mathbf{V} \mathbf{G}_\pi')^* (\mathbf{G}_\pi \mathbf{V}) \quad (11)$$

which is estimated by replacing π by $\hat{\pi}$. (See Matthews and Crowther (1995) for more on ML estimation procedure.)

3.2 Distribution fitting of grouped response vectors

Simultaneous distribution fitting to each response vector utilizing the ML estimation procedure above, entails that the cumulative relative frequencies in (8) should equal the cumulative distribution curve at the upper class boundaries in (2), for every cell in the multifactor design, i.e.

$$F(\mathbf{x}, \beta) = \begin{pmatrix} F_1(\mathbf{x}, \beta_1) \\ F_2(\mathbf{x}, \beta_2) \\ \vdots \\ F_T(\mathbf{x}, \beta_T) \end{pmatrix} = \begin{pmatrix} \pi_1 \\ \pi_2 \\ \vdots \\ \pi_T \end{pmatrix} \quad (12)$$

where

$$\beta' = (\beta'_1 \quad \beta'_2 \quad \cdots \quad \beta'_T) \quad (13)$$

is the concatenated vector of parameters for the T distribution functions to be estimated.

Consider the case where (12) may be transformed into the linear model

$$h(\pi) = \begin{pmatrix} h(\pi_1) \\ h(\pi_2) \\ \vdots \\ h(\pi_T) \end{pmatrix} = \begin{pmatrix} \mathbf{X}\alpha_1 \\ \mathbf{X}\alpha_2 \\ \vdots \\ \mathbf{X}\alpha_T \end{pmatrix} = (\mathbf{I}_T \otimes \mathbf{X}) \alpha \quad (14)$$

where

$$\alpha' = (\alpha'_1 \quad \alpha'_2 \quad \cdots \quad \alpha'_T) \quad (15)$$

and the α_t 's in (15) are functions of the β_t 's in (13) for $t = 1, 2, \dots, T$. According to (14), π is to be estimated such that $h(\pi_t)$ for $t = 1, 2, \dots, T$ should be in the vector space of $\mathbf{X}$ which implies the vector of constraints

$$\mathbf{g}_{\text{distr}}(\boldsymbol{\pi}) = \mathbf{0},$$

$$\mathbf{g}_{\text{distr}}(\boldsymbol{\pi}) = \begin{pmatrix} \mathbf{Q}_X h(\boldsymbol{\pi}_1) \\ \mathbf{Q}_X h(\boldsymbol{\pi}_2) \\ \vdots \\ \mathbf{Q}_X h(\boldsymbol{\pi}_T) \end{pmatrix} = (\mathbf{I}_T \otimes \mathbf{Q}_X) h(\boldsymbol{\pi}) \quad (16)$$

with $\mathbf{Q}_X = \mathbf{X}(\mathbf{X}'\mathbf{X})^* \mathbf{X}'$, the projection matrix orthogonal to $\mathbf{X}$ and $(\mathbf{X}'\mathbf{X})^*$ a generalized inverse of $(\mathbf{X}'\mathbf{X})$.

3.3 Explanatory variables in a multifactor model

Consider a descriptive measure, $\psi_t = \psi(\boldsymbol{\pi}_t)$ of the distributions $F_t(\mathbf{x}, \boldsymbol{\beta}_t)$ for $t = 1, 2, \dots, T$. This could be a mean, quartile, variance or any other measure. To explain the effect of the explanatory variables in the multifactor design on the response variable, the multifactor model

$$\boldsymbol{\psi} = \mathbf{Y}\boldsymbol{\gamma} : T \times 1 \quad (17)$$

where $\boldsymbol{\psi}$ is the vector of descriptive measures, $\mathbf{Y}$ is the design matrix and $\boldsymbol{\gamma}$ is the vector of parameters, is formulated. This formulation implies the additional constraints $\mathbf{g}_{\text{mod}}(\boldsymbol{\pi}) = \mathbf{0}$,

$$\mathbf{g}_{\text{mod}}(\boldsymbol{\pi}) = \mathbf{Q}_Y \boldsymbol{\psi} \quad (18)$$

with $\mathbf{Q}_Y = \mathbf{I} - \mathbf{Y}(\mathbf{Y}'\mathbf{Y})^* \mathbf{Y}'$, the projection matrix orthogonal to the columns of $\mathbf{Y}$ and $(\mathbf{Y}'\mathbf{Y})^*$ a generalized inverse of $(\mathbf{Y}'\mathbf{Y})$.

Employing the ML estimation process with constraints $\mathbf{g}(\boldsymbol{\pi}) = \mathbf{0}$,

$$\mathbf{g}(\boldsymbol{\pi}) = \begin{pmatrix} \mathbf{g}_{\text{distr}}(\boldsymbol{\pi}) \\ \mathbf{g}_{\text{mod}}(\boldsymbol{\pi}) \end{pmatrix} \quad (19)$$

the ML estimate $\hat{\boldsymbol{\pi}}$ has the following properties:

1. $\hat{\boldsymbol{\pi}}_t$ follows the cumulative distribution curve $F_t(\mathbf{x}, \boldsymbol{\beta}_t)$ for $t = 1, 2, \dots, T$ at the upper class limits of $\mathbf{x}$ and
2. $\hat{\psi}_t = \psi(\hat{\boldsymbol{\pi}}_t)$ for $t = 1, 2, \dots, T$ satisfy the multifactor model in (17).

It is now possible to evaluate the effect of the explanatory variables in the multifactor model by means of the ML estimate

$$\hat{\gamma} = (\mathbf{Y}'\mathbf{Y})^* \mathbf{Y}'\hat{\psi}.$$

4. Normal distribution fitting

4.1 Distribution fitting of response vectors

To fit normal distributions $N(\mu_t, \sigma_t^2)$ for $t = 1, 2, \dots, T$ simultaneously to the T cells of a multifactor design it is required that

$$\Phi(\mathbf{z}_t) = \boldsymbol{\pi}_t \quad (20)$$

where $\Phi(\cdot)$ is the vector valued cumulative distribution function of the standardized normal distribution and

$$\mathbf{z}_t = \frac{\mathbf{x} - \mu_t \mathbf{1}}{\sigma_t} \quad (21)$$

is the vector of standardized upper class limits for the t -th cell of the multifactor design.

Taking the inverse normal function from (20) leads to the linear model

$$\begin{aligned} \Phi^{-1}(\boldsymbol{\pi}_t) &= \frac{\mathbf{x} - \mu_t \mathbf{1}}{\sigma_t} \\ &= \mathbf{X}\boldsymbol{\alpha}_t \end{aligned} \quad (22)$$

where

$$\mathbf{X} = (\mathbf{x} \quad -\mathbf{1}) \quad (23)$$

is the design matrix for normality and

$$\boldsymbol{\alpha}_t = \begin{pmatrix} \alpha_{1t} \\ \alpha_{2t} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sigma_t} \\ \frac{\mu_t}{\sigma_t} \end{pmatrix} \quad (24)$$

is the vector of parameters for $t = 1, 2, \dots, T$.

To fit the T normal distributions simultaneously to the T frequency distributions it follows from (22) that it is required that

$$\Phi^{-1}(\boldsymbol{\pi}) = (\mathbf{I}_T \otimes \mathbf{X}) \boldsymbol{\alpha} \quad (25)$$

leading to the constraints for normality $\mathbf{g}_{\text{nor}}(\boldsymbol{\pi}) = \mathbf{0}$,

$$\mathbf{g}_{\text{nor}}(\boldsymbol{\pi}) = (\mathbf{I}_T \otimes \mathbf{Q}_X) \cdot \Phi^{-1}(\boldsymbol{\pi}) \quad (26)$$

with matrix of partial derivatives

$$\begin{aligned} \mathbf{G}_{\text{nor}}(\boldsymbol{\pi}) &= \frac{\partial \mathbf{g}_{\text{nor}}(\boldsymbol{\pi})}{\partial \boldsymbol{\pi}} \\ &= (\mathbf{I}_T \otimes \mathbf{Q}_X) \cdot \mathbf{D}_{\pi} \end{aligned}$$

where $\mathbf{D}_{\pi} = \frac{\partial \Phi^{-1}(\boldsymbol{\pi})}{\partial \boldsymbol{\pi}} = (\text{diag} [\phi(\Phi^{-1}(\boldsymbol{\pi}))])^{-1}$ and $\phi(\cdot)$ is the vector valued probability distribution function of the standardized normal distribution.

The ML estimate of $\boldsymbol{\alpha}$ follows from (25)

$$\hat{\boldsymbol{\alpha}} = (\mathbf{I}_T \otimes (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}') \cdot \Phi^{-1}(\hat{\boldsymbol{\pi}}) \quad (27)$$

which consists of two sets of estimators namely

$$\hat{\boldsymbol{\alpha}}_1 = \begin{pmatrix} \hat{\alpha}_{11} \\ \hat{\alpha}_{12} \\ \vdots \\ \hat{\alpha}_{1T} \end{pmatrix} = \begin{pmatrix} 1/\hat{\sigma}_1 \\ 1/\hat{\sigma}_2 \\ \vdots \\ 1/\hat{\sigma}_T \end{pmatrix} = (\mathbf{I}_T \otimes [(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}']_1) \cdot \Phi^{-1}(\hat{\boldsymbol{\pi}}) \quad (28)$$

and

$$\hat{\boldsymbol{\alpha}}_2 = \begin{pmatrix} \hat{\alpha}_{21} \\ \hat{\alpha}_{22} \\ \vdots \\ \hat{\alpha}_{2T} \end{pmatrix} = \begin{pmatrix} \hat{\mu}_1/\hat{\sigma}_1 \\ \hat{\mu}_2/\hat{\sigma}_2 \\ \vdots \\ \hat{\mu}_T/\hat{\sigma}_T \end{pmatrix} = (\mathbf{I}_T \otimes [(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}']_2) \cdot \Phi^{-1}(\hat{\boldsymbol{\pi}}) \quad (29)$$

where $[(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}']_i$ refers to the i -th row of the matrix $(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'$. The asymptotic covariance matrix of $\hat{\boldsymbol{\alpha}}$ follows from the Multivariate Delta

Theorem

$$\begin{aligned} \text{Cov}(\hat{\alpha}) &\cong \left(\frac{\partial \alpha}{\partial \pi} \right) \text{Cov}(\hat{\pi}) \left(\frac{\partial \alpha}{\partial \pi} \right)' \\ &= \left\{ \left(\mathbf{I}_T \otimes (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \right) \mathbf{D}_\pi \right\} \\ &\quad \text{Cov}(\hat{\pi}) \left\{ \left(\mathbf{I}_T \otimes (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \right) \mathbf{D}_\pi \right\}'. \end{aligned} \quad (30)$$

The ML estimates of μ and σ are obtained from

$$\hat{\mu} = \begin{pmatrix} \hat{\mu}_1 \\ \hat{\mu}_2 \\ \vdots \\ \hat{\mu}_T \end{pmatrix} = \begin{pmatrix} \hat{\alpha}_{21}/\hat{\alpha}_{11} \\ \hat{\alpha}_{22}/\hat{\alpha}_{12} \\ \vdots \\ \hat{\alpha}_{2T}/\hat{\alpha}_{1T} \end{pmatrix} = \hat{\alpha}_2/\hat{\alpha}_1 \quad (31)$$

and

$$\hat{\sigma} = \begin{pmatrix} \hat{\sigma}_1 \\ \hat{\sigma}_2 \\ \vdots \\ \hat{\sigma}_T \end{pmatrix} = \begin{pmatrix} 1/\hat{\alpha}_{11} \\ 1/\hat{\alpha}_{12} \\ \vdots \\ 1/\hat{\alpha}_{1T} \end{pmatrix} = \mathbf{1}/\hat{\alpha}_1 \quad (32)$$

where an element wise division for $\hat{\alpha}_2/\hat{\alpha}_1$ and $\mathbf{1}/\hat{\alpha}_1$ are understood in (31) and (32). The asymptotic covariance matrices for $\hat{\mu}$ and $\hat{\sigma}$ are

$$\text{Cov}(\hat{\mu}) \cong \mathbf{B}_\mu \text{Cov}(\hat{\alpha}) \mathbf{B}_\mu', \quad \mathbf{B}_\mu = \begin{pmatrix} \alpha_2 \\ \alpha_1^2 \end{pmatrix} \otimes (1 \ 0) + \begin{pmatrix} 1 \\ \alpha_1 \end{pmatrix} \otimes (0 \ 1) \quad (33)$$

and

$$\text{Cov}(\hat{\sigma}) \cong \mathbf{B}_\sigma \text{Cov}(\hat{\alpha}) \mathbf{B}_\sigma', \quad \mathbf{B}_\sigma = \begin{pmatrix} 1 \\ \alpha_1^2 \end{pmatrix} \otimes (1 \ 0) \quad (34)$$

respectively.

4.2 Equality of variances

Equality of variances may be formulated by

$$\mathbf{H}\alpha_1 = \mathbf{0} \quad (35)$$

where

$$\mathbf{H} = \begin{pmatrix} 1 & -1 & 0 & \cdots & 0 \\ 1 & 0 & -1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \cdots & -1 \end{pmatrix} = \begin{pmatrix} \mathbf{1}_{(T-1)} & -\mathbf{I}_{(T-1)} \end{pmatrix}$$

is a $((T-1) \times T)$ matrix of contrasts and

$$\boldsymbol{\alpha}_1 = \begin{pmatrix} \alpha_{11} \\ \alpha_{12} \\ \vdots \\ \alpha_{1T} \end{pmatrix} = \begin{pmatrix} 1/\sigma_1 \\ 1/\sigma_2 \\ \vdots \\ 1/\sigma_T \end{pmatrix} = \left(\mathbf{I}_T \otimes [(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}']_1 \right) \Phi^{-1}(\boldsymbol{\pi}) .$$

Hence, the constraints for equality of variances is $\mathbf{g}_{\text{var}}(\boldsymbol{\pi}) = \mathbf{0}$,

$$\mathbf{g}_{\text{var}}(\boldsymbol{\pi}) = \mathbf{H} \left(\mathbf{I}_T \otimes [(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}']_1 \right) \Phi^{-1}(\boldsymbol{\pi}) . \quad (36)$$

Implementing the vector of constraints

$$\mathbf{g}(\boldsymbol{\pi}) = \begin{pmatrix} \mathbf{g}_{\text{nor}}(\boldsymbol{\pi}) \\ \mathbf{g}_{\text{var}}(\boldsymbol{\pi}) \end{pmatrix} = \begin{pmatrix} (\mathbf{I}_T \otimes \mathbf{Q}_X) \\ \mathbf{H} \left(\mathbf{I}_T \otimes [(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}']_1 \right) \end{pmatrix} \Phi^{-1}(\boldsymbol{\pi})$$

in the ML estimation procedure yields the ML estimate of $\boldsymbol{\pi}$ such that:

1. $\hat{\boldsymbol{\pi}}_t, (t = 1, 2, \dots, T)$ follows a cumulative normal distribution curve at the upper boundaries of $\mathbf{x}$ and
2. the fitted normal distributions have equal variances over the T cells.

4.3 Multifactor model

Under equality of variances, discussed in the previous section, the ML estimate of $\boldsymbol{\pi}$ may be estimated under the additional constraints that the means of the fitted normal distributions follow a specific design i.e.

$$\boldsymbol{\mu} = \mathbf{Y}\boldsymbol{\gamma} . \quad (37)$$

Since estimation takes place under equality of variances the vector of parameters

$$\boldsymbol{\alpha}_2 = \begin{pmatrix} \alpha_{21} \\ \alpha_{22} \\ \vdots \\ \alpha_{2T} \end{pmatrix} = \begin{pmatrix} \mu_1/\sigma \\ \mu_2/\sigma \\ \vdots \\ \mu_T/\sigma \end{pmatrix}$$

may be referred to as the so-called standardized effects. It follows from (37) that $\boldsymbol{\pi}$ is to be estimated such that the fitted means, $\hat{\mu}_t$ for $t = 1, 2, \dots, T$, are in the vector space of $\mathbf{Y}$ implying that the additional constraints $\mathbf{g}_{\text{mod}}(\boldsymbol{\pi}) = \mathbf{0}$,

$$\mathbf{g}_{\text{mod}}(\boldsymbol{\pi}) = \mathbf{Q}_Y \boldsymbol{\alpha}_2 \quad (38)$$

are to be implemented in the vector of constraints

$$\mathbf{g}(\boldsymbol{\pi}) = \begin{pmatrix} \mathbf{g}_{\text{nor}}(\boldsymbol{\pi}) \\ \mathbf{g}_{\text{var}}(\boldsymbol{\pi}) \\ \mathbf{g}_{\text{mod}}(\boldsymbol{\pi}) \end{pmatrix} = \begin{pmatrix} (\mathbf{I}_T \otimes \mathbf{Q}_X) \\ \mathbf{H} \left(\mathbf{I}_T \otimes \left[(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \right]_1 \right) \\ \mathbf{Q}_Y \left(\mathbf{I}_T \otimes \left[(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \right]_2 \right) \end{pmatrix} \cdot \Phi^{-1}(\boldsymbol{\pi}) . \quad (39)$$

The ML estimate of $\boldsymbol{\pi}$ under the constraints (39) will now be estimated such that

1. $\hat{\pi}_t$ for $t = 1, 2, \dots, T$ follows a cumulative normal distribution curve at the upper boundaries of $\mathbf{x}$,
2. the fitted normal distributions have equal variances and
3. the ML estimate $\hat{\boldsymbol{\mu}}$ satisfies the multifactor design in (37).

4.4 Application: Single-factor model

In this application the effect of achievement in Grade 12 Mathematics (MATHS) on the performance of first year Statistics (STATS) is to be investigated. The STATS exam paper counted out of 108 marks and the results were classified into a total of 5 categories to illustrate the technique. The data is summarized in Table 2.

Table 2 Data set of 898 first year students

MATHS	STATS					Total
	[0 – 39]	[40 – 49]	[50 – 59]	[60 – 74]	[75 – 108]	
A	0	4	19	53	84	160
B	3	17	35	65	19	139
C	24	44	56	68	19	211
D	43	57	82	48	6	236
E	59	53	26	13	1	152
Total	129	175	218	247	129	898

The performance of STATS is to be evaluated over the 5 levels of MATHS resulting in a single-factor design with $T = 5$ cells. The response variable STATS is categorised into $k = 5$ class intervals with upper class limits $\mathbf{x} = (39.5 \ 49.5 \ 59.5 \ 74.5)$. Since the exam mark is treated as a continuous variable and recorded to the nearest integer, the upper class limits in $\mathbf{x}$ are taken half-way between the gaps of the respective class intervals.

In this example a single factor model, with an ordinal linear trend in STATS over the levels of MATHS is to be fitted, implying the single-factor model

$$\begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \mu_4 \\ \mu_5 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 1 & 1 \\ 1 & 0 \\ 1 & -1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} \gamma_0 \\ \gamma_1 \end{pmatrix} = \mathbf{Y}\boldsymbol{\gamma} \quad (40)$$

leading to the vector of constraints

$$\mathbf{g}_{\text{mod}}(\boldsymbol{\pi}) = \mathbf{Q}_Y \boldsymbol{\gamma} \quad (41)$$

where $\mathbf{Q}_Y = \mathbf{Y}(\mathbf{Y}'\mathbf{Y})^{-1}\mathbf{Y}'$. Utilizing the ML estimation procedure with vector of constraints

$$\mathbf{g}(\boldsymbol{\pi}) = \begin{pmatrix} \mathbf{g}_{\text{nor}}(\boldsymbol{\pi}) \\ \mathbf{g}_{\text{var}}(\boldsymbol{\pi}) \\ \mathbf{g}_{\text{mod}}(\boldsymbol{\pi}) \end{pmatrix} = \mathbf{0}$$

where $\mathbf{g}_{\text{nor}}(\boldsymbol{\pi})$ is defined in (26), $\mathbf{g}_{\text{var}}(\boldsymbol{\pi})$ is defined in (36) and $\mathbf{g}_{\text{mod}}(\boldsymbol{\pi})$ is defined in (41) the ML estimate of $\boldsymbol{\pi}$ leads to the estimated frequencies listed in Table 3.

Table 3 Estimated frequencies for single-factor model with ordinal explanatory variable

MATHS	STATS					Total
	[0–40)	[40–50)	[50–60)	[60–75)	[75–108]	
A	0.8	5	19.3	65.8	69.1	160
B	3.2	12.6	31.1	61	31.1	139
C	17	39.8	63	72.6	18.7	211
D	49.1	66.3	67.7	46.7	6.2	236
E	62.5	45.8	30.1	12.7	0.9	152

An adequate fit is evident when comparing the observed frequencies in Table 2 with the estimated frequencies in Table 3, which is further supported by the Pearson goodness of fit statistic $\chi^2 = 25$ ($df = 17$, the number of linear independent constraints in $\mathbf{g}(\boldsymbol{\pi}) = \mathbf{0}$). The estimated normal distributions, with ML estimates are displayed in Table 4.

The ML estimate of $\boldsymbol{\gamma}$ in the multifactor model (40)

$$\hat{\boldsymbol{\gamma}} = (\mathbf{Y}'\mathbf{Y}^{-1}) \mathbf{Y}'\hat{\boldsymbol{\mu}} = \begin{pmatrix} \hat{\gamma}_0 \\ \hat{\gamma}_1 \end{pmatrix} = \begin{pmatrix} 57.3 \\ 7.5 \end{pmatrix}$$

indicates that the estimated average STATS mark for students with a C symbol for MATHS is 57.3 and that every increase of one symbol in MATHS implies an estimated increase of 7.5 marks in STATS. The standard errors of $\hat{\boldsymbol{\gamma}}$

$$\hat{\sigma}_{\hat{\boldsymbol{\gamma}}} = \begin{pmatrix} \hat{\sigma}_{\hat{\gamma}_0} \\ \hat{\sigma}_{\hat{\gamma}_1} \end{pmatrix} = \begin{pmatrix} 0.4563 \\ 0.3521 \end{pmatrix}$$

enable the construction of confidence intervals and the testing of relevant hypotheses. The mean in the i -th level of MATHS may be expressed in terms of the single factor model

$$\mu_i = \tau_0 + \tau_i^M \quad i = 1, 2, \dots, 5$$

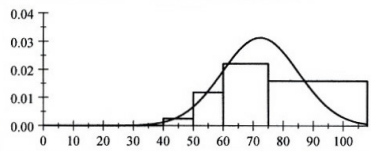
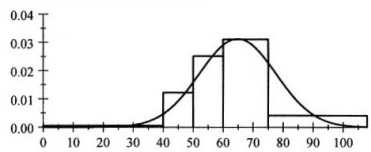
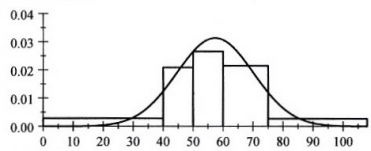
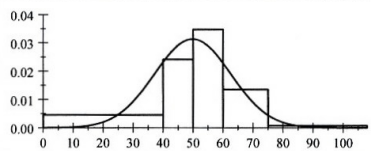
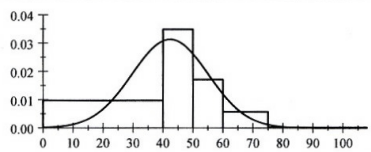
where

$$\tau_0 = \text{overall mean}$$

$$\tau_i^M = \text{effect for the } i\text{-th level of MATHS} \quad i = 1, 2, \dots, 5$$

The ML estimates for these single factor model effects, together with their standard errors (indicated in brackets) are also tabulated in Table 4.

Table 4 ML estimates for single-factor model with ordinal explanatory variable

MATHS	STATS	n	$\hat{\mu}$ ($\hat{\sigma}_{\hat{\mu}}$)	$\hat{\sigma}$ ($\hat{\sigma}_{\hat{\sigma}}$)	$\hat{\tau}^M$ ($\hat{\sigma}_{\hat{\tau}^M}$)
A		160	72.3 (0.87)	12.7 (0.40)	15.0 (0.70)
B		139	64.8 (0.60)	12.7 (0.40)	7.5 (0.35)
C		211	57.3 (0.46)	12.7 (0.40)	0.0 (0.00)
D		236	49.8 (0.55)	12.7 (0.40)	-7.5 (0.35)
E		152	42.4 (0.81)	12.7 (0.40)	-15.0 (0.70)
$\hat{\tau}_0$ ($\hat{\sigma}_{\hat{\tau}_0}$)					57.3 (0.46)

5. Log-logistic distribution fitting

In the case where the grouped response vector has a positive skew distribution, the log-logistic distribution with probability distribution function

$$f(x; \kappa, \theta) = \frac{e^{\theta \kappa x^{\kappa-1}}}{(1 + e^{\theta x^{\kappa}})^2}$$

may be fitted effectively to the T frequency distributions in a multifactor design. Due to the skewness of the response variable, the median of the fitted log-logistic distributions will be used as a representative measure for each of the T frequency distributions.

From the cumulative distribution function of the log-logistic distribution

$$F(x; \kappa, \theta) = \frac{e^{\theta x^{\kappa}}}{1 + e^{\theta x^{\kappa}}}$$

the median ν is obtained from

$$\frac{e^{\theta \nu^{\kappa}}}{1 + e^{\theta \nu^{\kappa}}} = 0.5$$

leading to

$$\nu = \exp \left(-\frac{\theta}{\kappa} \right). \quad (42)$$

In the multifactor model the medians will be employed in a linear model to determine the effect of the explanatory variables on the grouped response variable.

5.1 Distribution fitting of response vectors

Setting $F(\mathbf{x}; \kappa, \theta) = \pi_t$ for $t = 1, 2, \dots, T$ and using elementwise division it follows that

$$\frac{e^{\theta \nu^{\kappa}} \mathbf{1}}{1 + e^{\theta \nu^{\kappa}} \mathbf{1}} = \boldsymbol{\pi}_t$$

and therefore

$$\frac{\boldsymbol{\pi}_t}{1 - \boldsymbol{\pi}_t} = e^{\theta_t \mathbf{x}^{\kappa_t}}.$$

Taking the natural logarithm leads to the linear model

$$\begin{aligned} \ln \left(\frac{\pi_t}{1 - \pi_t} \right) &= \kappa_t \ln \mathbf{x} + \theta_t \mathbf{1} \\ &= \mathbf{X} \boldsymbol{\alpha}_t \end{aligned} \quad (43)$$

where

$$\mathbf{X} = \begin{pmatrix} \ln \mathbf{x} & \mathbf{1} \end{pmatrix} \quad (44)$$

is the design matrix and

$$\boldsymbol{\alpha}_t = \begin{pmatrix} \alpha_{1t} \\ \alpha_{2t} \end{pmatrix} = \begin{pmatrix} \kappa \\ \theta \end{pmatrix} \quad (45)$$

is the vector of parameters for $t = 1, 2, \dots, T$. To fit log-logistic distributions it follows from (43) that $\hat{\pi}$ is to be estimated such that $\ln \left(\frac{\hat{\pi}_t}{1 - \hat{\pi}_t} \right)$ should be in the vector space of $\mathbf{X} = \begin{pmatrix} \ln \mathbf{x} & \mathbf{1} \end{pmatrix}$ for $t = 1, 2, \dots, T$.

Stacking the linear functions in (43) for $t = 1, 2, \dots, T$ below each other leads to the concatenated vector

$$\ln \left(\frac{\boldsymbol{\pi}}{\mathbf{1} - \boldsymbol{\pi}} \right) = (\mathbf{I}_T \otimes \mathbf{X}) \boldsymbol{\alpha} \quad (46)$$

and suggests the vector of constraints $\mathbf{g}_{\log}(\boldsymbol{\pi}) = \mathbf{0}$,

$$\mathbf{g}_{\log}(\boldsymbol{\pi}) = (\mathbf{I}_T \otimes \mathbf{Q}_X) \cdot \ln \left(\frac{\boldsymbol{\pi}}{\mathbf{1} - \boldsymbol{\pi}} \right) \quad (47)$$

with matrix of partial derivatives

$$\mathbf{G}_{\log}(\boldsymbol{\pi}) = (\mathbf{I}_T \otimes \mathbf{Q}_X) \cdot \mathbf{D}_{\pi}$$

where $\mathbf{D}_{\pi} = \{\text{diag}(\boldsymbol{\pi})\}^{-1} + \{\text{diag}(\mathbf{1} - \boldsymbol{\pi})\}$.

The ML estimate of $\boldsymbol{\alpha}$ follows from the ML estimate of $\boldsymbol{\pi}$ under constraints

$$\hat{\boldsymbol{\alpha}} = \left(\mathbf{I}_T \otimes (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \right) \cdot \ln \left(\frac{\hat{\boldsymbol{\pi}}}{\mathbf{1} - \hat{\boldsymbol{\pi}}} \right) \quad (48)$$

which consists out of two sets of estimators namely

$$\hat{\boldsymbol{\kappa}} = \left(\mathbf{I}_T \otimes \left[(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \right]_1 \right) \cdot \ln \left(\frac{\hat{\boldsymbol{\pi}}}{\mathbf{1} - \hat{\boldsymbol{\pi}}} \right) \quad (49)$$

and

$$\hat{\boldsymbol{\theta}} = \left(\mathbf{I}_T \otimes \left[(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \right]_2 \right) \cdot \ln \left(\frac{\hat{\boldsymbol{\pi}}}{\mathbf{1} - \hat{\boldsymbol{\pi}}} \right). \quad (50)$$

The asymptotic covariance matrix of $\hat{\boldsymbol{\alpha}}$ is

$$\text{Cov}(\hat{\boldsymbol{\alpha}}) \cong \left\{ \left(\mathbf{I}_T \otimes (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \right) \mathbf{D}_\pi \right\} \text{Cov}(\hat{\boldsymbol{\pi}}) \left\{ \left(\mathbf{I}_T \otimes (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \right) \mathbf{D}_\pi \right\}'. \quad (51)$$

The asymptotic standard errors of $\hat{\boldsymbol{\kappa}}$ and $\hat{\boldsymbol{\theta}}$ can be calculated directly from

$$\begin{aligned} \text{Cov}(\hat{\boldsymbol{\kappa}}) &\cong \left\{ \left(\mathbf{I}_T \otimes \left[(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \right]_1 \right) \mathbf{D}_\pi \right\} \\ &\quad \text{Cov}(\hat{\boldsymbol{\pi}}) \left\{ \left(\mathbf{I}_T \otimes \left[(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \right]_1 \right) \mathbf{D}_\pi \right\}' \end{aligned} \quad (52)$$

and

$$\begin{aligned} \text{Cov}(\hat{\boldsymbol{\theta}}) &\cong \left\{ \left(\mathbf{I}_T \otimes \left[(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \right]_2 \right) \mathbf{D}_\pi \right\} \\ &\quad \text{Cov}(\hat{\boldsymbol{\pi}}) \left\{ \left(\mathbf{I}_T \otimes \left[(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \right]_2 \right) \mathbf{D}_\pi \right\}'. \end{aligned} \quad (53)$$

5.2 Multifactor model

The medians (42) of the fitted log-logistic distributions will be employed in a linear model to evaluate the effect of the explanatory variables on the response variable over the T cells of the multifactor design. The concatenated vector of medians for the T cells is

$$\boldsymbol{\nu} = \begin{pmatrix} \nu_1 \\ \nu_2 \\ \vdots \\ \nu_T \end{pmatrix} = \exp \begin{pmatrix} \begin{pmatrix} -\frac{\theta_1}{\kappa_1} \\ -\frac{\theta_2}{\kappa_2} \\ \vdots \\ -\frac{\theta_T}{\kappa_T} \end{pmatrix} \end{pmatrix} = \exp \left(-\frac{\boldsymbol{\theta}}{\boldsymbol{\kappa}} \right).$$

Let

$$\boldsymbol{\nu} = \mathbf{Y}\boldsymbol{\gamma} \quad (54)$$

specify the multifactor model. The objective is to estimate $\boldsymbol{\pi}$ such that $\boldsymbol{\nu}$ is in the vector space of $\mathbf{Y}$ implying the vector of constraints $\mathbf{g}_{\text{mod}}(\boldsymbol{\pi}) = \mathbf{0}$,

$$\mathbf{g}_{\text{mod}}(\boldsymbol{\pi}) = \mathbf{Q}_Y \boldsymbol{\nu}. \quad (55)$$

Implementing the chain rule, the matrix of partial derivatives

$$\mathbf{G}_{\text{mod}}(\boldsymbol{\pi}) = \mathbf{Q}_Y \mathbf{A} \left(\mathbf{I}_T \otimes (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \right) \mathbf{D}_{\pi}$$

follows, where

$$\mathbf{A} = \frac{\partial \boldsymbol{\nu}}{\partial \boldsymbol{\alpha}} = \begin{pmatrix} \mathbf{A}_1 & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_2 & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{A}_T \end{pmatrix} \quad (56)$$

with

$$\mathbf{A}_t = \frac{\partial \boldsymbol{\nu}_t}{\partial \boldsymbol{\alpha}_t} = \begin{pmatrix} \frac{\theta_t}{\kappa_t^2} \exp\left(-\frac{\theta_t}{\kappa_t}\right) & -\frac{1}{\kappa_t} \exp\left(-\frac{\theta_t}{\kappa_t}\right) \end{pmatrix}, \quad t = 1, 2, \dots, T.$$

Employing the ML estimation procedure with

$$\mathbf{g}(\boldsymbol{\pi}) = \begin{pmatrix} \mathbf{g}_{\log}(\boldsymbol{\pi}) \\ \mathbf{g}_{\text{mod}}(\boldsymbol{\pi}) \end{pmatrix} \quad \text{and} \quad \mathbf{G}_{\pi} = \begin{pmatrix} \mathbf{G}_{\log}(\boldsymbol{\pi}) \\ \mathbf{G}_{\text{mod}}(\boldsymbol{\pi}) \end{pmatrix}$$

leads to the ML estimate of $\boldsymbol{\pi}$ under constraints such that:

1. the elements of $\hat{\boldsymbol{\pi}}_1, \hat{\boldsymbol{\pi}}_2, \dots, \hat{\boldsymbol{\pi}}_T$ follow T log-logistic curves at the upper class limits $\mathbf{x}$ and
2. the ML estimate $\hat{\boldsymbol{\nu}}$ is a linear combination of $\mathbf{Y}$ in (54).

The asymptotic covariance matrix of $\hat{\boldsymbol{\nu}}$ is

$$\text{Cov}(\hat{\boldsymbol{\nu}}) \approx \mathbf{A} \text{Cov}(\hat{\boldsymbol{\alpha}}) \mathbf{A}' \quad (57)$$

with $\mathbf{A}$ given in (56).

The multifactor model (54) can be explained by the ML estimate

$$\hat{\boldsymbol{\gamma}} = (\mathbf{Y}'\mathbf{Y})^* \mathbf{Y}'\hat{\boldsymbol{\nu}} \quad (58)$$

and for the purpose of statistical inference, the standard errors are obtained from the asymptotic covariance matrix

$$\text{Cov}(\hat{\gamma}) \cong \{(\mathbf{Y}'\mathbf{Y})^* \mathbf{Y}'\} \text{Cov}(\hat{\nu}) \{(\mathbf{Y}'\mathbf{Y})^* \mathbf{Y}'\}' . \quad (59)$$

5.3 Application: Two-factor model

The premiums (PREMIUM) of 8334 policyholders in the short-term insurance are classified into the 5 categories listed in Table 5.

Table 5 Frequency distribution of PREMIUM

PREMIUM	Frequency
R51-R200	1920
R201-R300	2726
R301-R400	1677
R401-R500	930
R500-R1000	1081

The objective is to explain the effect of the two explanatory variables, the age of the policyholder (AGE) and the type of product (PRODUCT) on the PREMIUM of the policy. The variable AGE is classified into 4 categories, while PRODUCT consists out of three types of insurance policies. A cross classification of these two factors results in a two-factor design with a total of 12 cells, each containing a frequency distribution with sample sizes summarized in Table 6.

Table 6 Contingency table of AGE and PRODUCT

AGE	PRODUCT			Total
	I	II	III	
20-29	930	415	461	1806
30-39	1105	800	1017	2922
40-49	832	764	656	2252
50-59	448	416	490	1354
Total	3315	2395	2624	8334

The 12 cells in Table 6 can be modeled by a two-factor model. Due to the positive skew nature of PREMIUM, log-logistic curves will be fitted

simultaneously to the 12 frequency distributions in the two-factor design. The variable PREMIUM is modeled in hundreds of rands, which implies that the vector of upper class boundaries is $\mathbf{x} = (2 \ 3 \ 4 \ 5)$. The variable AGE with class midpoints $\mathbf{z}_A = (24.5 \ 34.5 \ 44.5 \ 54.5)$ will be introduced as a covariate in a two-factor model enabling different slopes for PREMIUM on AGE for each PRODUCT leading to the two-factor model

$$\boldsymbol{\nu} = \mathbf{Y}\boldsymbol{\gamma} = \begin{pmatrix} 1 & 24.5 & 1 & 0 & 24.5 & 0 \\ 1 & 24.5 & 0 & 1 & 0 & 24.5 \\ 1 & 24.5 & -1 & -1 & -24.5 & -24.5 \\ 1 & 34.5 & 1 & 0 & 34.5 & 0 \\ 1 & 34.5 & 0 & 1 & 0 & 34.5 \\ 1 & 34.5 & -1 & -1 & -34.5 & -34.5 \\ 1 & 44.5 & 1 & 0 & 44.5 & 0 \\ 1 & 44.5 & 0 & 1 & 0 & 44.5 \\ 1 & 44.5 & -1 & -1 & -44.5 & -44.5 \\ 1 & 54.5 & 1 & 0 & 54.5 & 0 \\ 1 & 54.5 & 0 & 1 & 0 & 54.5 \\ 1 & 54.5 & -1 & -1 & -54.5 & -54.5 \end{pmatrix} \begin{pmatrix} \gamma_0 \\ \gamma_1 \\ \gamma_2 \\ \gamma_3 \\ \gamma_4 \\ \gamma_5 \end{pmatrix}. \quad (60)$$

The vector of constraints to be imposed in the ML estimation procedure is

$$\mathbf{g}(\boldsymbol{\pi}) = \begin{pmatrix} \mathbf{g}_{\log}(\boldsymbol{\pi}) \\ \mathbf{g}_{\text{mod}}(\boldsymbol{\pi}) \end{pmatrix} = \mathbf{0} \quad (61)$$

where $\mathbf{g}_{\log}(\boldsymbol{\pi})$ is defined in (47) and $\mathbf{g}_{\text{mod}}(\boldsymbol{\pi})$ is defined in (55). The observed and estimated frequencies are tabulated in Table 7. The Pearson goodness of fit statistic is $\chi^2 = 49.9$ ($df = 30$) suggests an adequate fit.

The ML estimate for $\boldsymbol{\gamma}$ in the multifactor model (60) is

$$\hat{\boldsymbol{\gamma}} = (\mathbf{Y}'\mathbf{Y})^{-1} \mathbf{Y}'\hat{\boldsymbol{\nu}} = \begin{pmatrix} \hat{\gamma}_0 \\ \hat{\gamma}_1 \\ \hat{\gamma}_2 \\ \hat{\gamma}_3 \\ \hat{\gamma}_4 \\ \hat{\gamma}_5 \end{pmatrix} = \begin{pmatrix} 3.4879 \\ -0.01532 \\ -0.39227 \\ 0.33708 \\ 0.00447 \\ -0.01963 \end{pmatrix}$$

implying that the overall trend in PREMIUM over AGE is

$$\hat{\nu}_i = 3.4879 - 0.01532z_i^A.$$

Table 7 Observed frequencies (f) and expected frequencies (e)

AGE	PRODUCT								
	I			II			III		
20-29	Premium	f	e	Premium	f	e	Premium	f	e
	R51-R200	157	166	R51-R200	49	68	R51-R200	69	61
	R201-R300	371	359	R201-R300	148	144	R201-R300	108	108
	R301-R400	232	239	R301-R400	101	109	R301-R400	103	107
	R401-R500	93	96	R401-R500	62	51	R401-R500	81	73
	R500+	77	70	R500+	55	44	R500+	100	112
30-39	Premium	f	e	Premium	f	e	Premium	f	e
	R51-R200	267	271	R51-R200	213	194	R51-R200	132	145
	R201-R300	400	378	R201-R300	328	317	R201-R300	249	235
	R301-R400	212	237	R301-R400	147	174	R301-R400	211	226
	R401-R500	115	109	R401-R500	71	66	R401-R500	156	155
	R500+	111	109	R500+	41	49	R500+	269	256
40-49	Premium	f	e	Premium	f	e	Premium	f	e
	R51-R200	241	229	R51-R200	302	289	R51-R200	73	84
	R201-R300	278	289	R201-R300	275	283	R201-R300	168	155
	R301-R400	167	167	R301-R400	117	117	R301-R400	151	155
	R401-R500	84	74	R401-R500	40	42	R401-R500	94	105
	R500+	62	72	R500+	30	33	R500+	170	156
50-59	Premium	f	e	Premium	f	e	Premium	f	e
	R51-R200	135	139	R51-R200	205	223	R51-R200	77	73
	R201-R300	150	155	R201-R300	130	120	R201-R300	121	112
	R301-R400	89	83	R301-R400	49	43	R301-R400	98	106
	R401-R500	40	36	R401-R500	18	16	R401-R500	76	73
	R500+	34	35	R500+	14	15	R500+	118	126

Due to the interaction that exists between AGE and PRODUCT, the three regression equations for PREMIUM are as follows:

PRODUCT I:

$$\hat{\nu}_{i1} = 3.0956 - 0.01085z_i^A$$

PRODUCT II:

$$\hat{\nu}_{i2} = 3.8250 - 0.03496z_i^A$$

PRODUCT III:

$$\hat{\nu}_{i3} = 3.5431 - 0.00016z_i^A$$

From the three regression equations above it follows that the estimated medians follow an unique trend over AGE for each of the three Products. For PRODUCT I the premium drops with an estimated R1.09 per year, for PRODUCT II the premium drops with an estimated R3.50 per year and for PRODUCT III no real linear trend over AGE is evident. The fitted log-logistic distributions are displayed in Table 8.

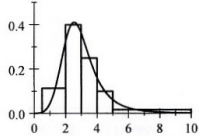
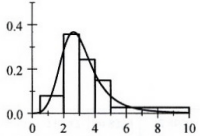
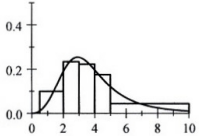
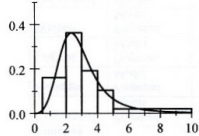
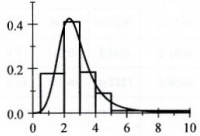
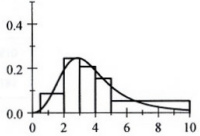
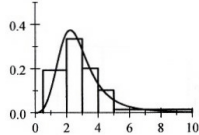
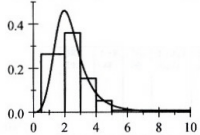
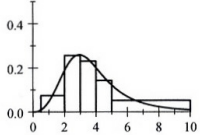
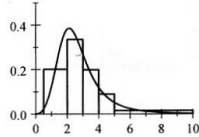
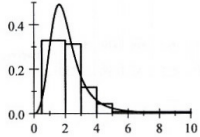
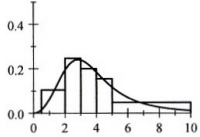
The median in the i -th level of AGE and the j -th level of PRODUCT can be expressed in terms of the two-factor model

$$\nu_{ij} = \tau_0 + \tau_i^A + \tau_j^P + \tau_{ij}^{AP}, \quad i = 1, 2, 3, 4 \quad \text{and} \quad j = 1, 2, 3$$

where

$$\begin{aligned} \tau_0 &= \bar{\nu} & : & \text{overall median} \\ \tau_i^A &= \bar{\nu}_i^A - \tau_0 & : & \text{effect for the } i\text{-th level of AGE} \\ \tau_j^P &= \bar{\nu}_j^P - \tau_0 & : & \text{effect for the } j\text{-th level of PRODUCT} \\ \tau_{ij}^{AP} &= \nu_{ij}^{AP} - (\tau_0 + \tau_i^A + \tau_j^P) & : & \text{interaction effect for the } i\text{-th level} \\ & & & \text{of AGE and } j\text{-th level of PRODUCT} \end{aligned}$$

Table 8: ML estimates for two-factor model with interaction and AGE treated as covariate

AGE	PRODUCT		
	P1	P2	P3
20-29	 $\hat{\kappa} = 4.409$ $\hat{\nu} = 2.830$ $\hat{\theta} = -4.587$ $\hat{\sigma}_{\hat{\nu}} = 0.033$	 $\hat{\kappa} = 4.110$ $\hat{\nu} = 2.969$ $\hat{\theta} = -4.472$ $\hat{\sigma}_{\hat{\nu}} = 0.046$	 $\hat{\kappa} = 3.286$ $\hat{\nu} = 3.539$ $\hat{\theta} = -4.153$ $\hat{\sigma}_{\hat{\nu}} = 0.068$
30-39	 $\hat{\kappa} = 3.644$ $\hat{\nu} = 2.721$ $\hat{\theta} = -3.648$ $\hat{\sigma}_{\hat{\nu}} = 0.022$	 $\hat{\kappa} = 4.213$ $\hat{\nu} = 2.619$ $\hat{\theta} = -4.056$ $\hat{\sigma}_{\hat{\nu}} = 0.027$	 $\hat{\kappa} = 3.152$ $\hat{\nu} = 3.538$ $\hat{\theta} = -3.983$ $\hat{\sigma}_{\hat{\nu}} = 0.042$
40-49	 $\hat{\kappa} = 3.621$ $\hat{\nu} = 2.613$ $\hat{\theta} = -3.477$ $\hat{\sigma}_{\hat{\nu}} = 0.028$	 $\hat{\kappa} = 3.921$ $\hat{\nu} = 2.269$ $\hat{\theta} = -3.214$ $\hat{\sigma}_{\hat{\nu}} = 0.026$	 $\hat{\kappa} = 3.356$ $\hat{\nu} = 3.536$ $\hat{\theta} = -4.238$ $\hat{\sigma}_{\hat{\nu}} = 0.044$
50-59	 $\hat{\kappa} = 3.566$ $\hat{\nu} = 2.504$ $\hat{\theta} = -3.273$ $\hat{\sigma}_{\hat{\nu}} = 0.045$	 $\hat{\kappa} = 3.468$ $\hat{\nu} = 1.920$ $\hat{\theta} = -2.262$ $\hat{\sigma}_{\hat{\nu}} = 0.043$	 $\hat{\kappa} = 3.051$ $\hat{\nu} = 3.534$ $\hat{\theta} = -3.852$ $\hat{\sigma}_{\hat{\nu}} = 0.072$

The effects of the two-factor model, with standard errors are tabulated in Table 9.

Table 9 Effects for the two-factor model

AGE	PRODUCT			Total
	I	II	III	
20-29	$\hat{\tau}^{AP} = -0.067$	$\hat{\tau}^{AP} = 0.294$	$\hat{\tau}^{AP} = -0.227$	$\hat{\tau}^A = 0.230$
	$\hat{\sigma}_{\hat{\tau}^{AP}} = 0.031$	$\hat{\sigma}_{\hat{\tau}^{AP}} = 0.034$	$\hat{\sigma}_{\hat{\tau}^{AP}} = 0.042$	$\hat{\sigma}_{\hat{\tau}^A} = 0.026$
30-39	$\hat{\tau}^{AP} = -0.022$	$\hat{\tau}^{AP} = 0.098$	$\hat{\tau}^{AP} = -0.076$	$\hat{\tau}^A = 0.077$
	$\hat{\sigma}_{\hat{\tau}^{AP}} = 0.010$	$\hat{\sigma}_{\hat{\tau}^{AP}} = 0.011$	$\hat{\sigma}_{\hat{\tau}^{AP}} = 0.014$	$\hat{\sigma}_{\hat{\tau}^A} = 0.009$
40-49	$\hat{\tau}^{AP} = 0.022$	$\hat{\tau}^{AP} = -0.098$	$\hat{\tau}^{AP} = 0.076$	$\hat{\tau}^A = -0.077$
	$\hat{\sigma}_{\hat{\tau}^{AP}} = 0.010$	$\hat{\sigma}_{\hat{\tau}^{AP}} = 0.011$	$\hat{\sigma}_{\hat{\tau}^{AP}} = 0.014$	$\hat{\sigma}_{\hat{\tau}^A} = 0.009$
50-59	$\hat{\tau}^{AP} = 0.067$	$\hat{\tau}^{AP} = -0.294$	$\hat{\tau}^{AP} = 0.227$	$\hat{\tau}^A = -0.230$
	$\hat{\sigma}_{\hat{\tau}^{AP}} = 0.031$	$\hat{\sigma}_{\hat{\tau}^{AP}} = 0.034$	$\hat{\sigma}_{\hat{\tau}^{AP}} = 0.042$	$\hat{\sigma}_{\hat{\tau}^A} = 0.026$
Total	$\hat{\tau}_P = -0.216$	$\hat{\tau}_P = -0.438$	$\hat{\tau}_P = 0.654$	$\hat{\tau}_0 = 2.883$
	$\hat{\sigma}_{\hat{\tau}_P} = 0.021$	$\hat{\sigma}_{\hat{\tau}_P} = 0.022$	$\hat{\sigma}_{\hat{\tau}_P} = 0.028$	$\hat{\sigma}_{\hat{\tau}_0} = 0.017$

Investigating the effects in Table 9, the marginal and the partial trends over AGE may be examined. Overall, the PREMIUM starts with R23 above the overall median of R288.30 and drops down linearly over the AGE categories to R23 below the overall median. It is interesting to note that this overall drop in PREMIUM seen by the $\hat{\tau}^A$ -values is cancelled out by the interaction effects, the $\hat{\tau}^{AP}$ -values for PRODUCT III, implying no linear trend over AGE for PRODUCT III. For PRODUCT II the effect of AGE on PREMIUM is rather drastic. Starting with an estimated R29.40 above the marginal effect for the youngest policy holders and dropping to R29.40 below the marginal effect for the oldest policy holders. From the marginal effects for PRODUCT it is clear that PRODUCT III is the most expensive product with an estimated R65.4 above the overall median.

6. Conclusion

The primary concern of this paper is to foster the basic theoretical concepts and methodology of how to model grouped data in a multifactor design. Although it is explained how to fit the normal distribution and the log-logistic distribution, any other distribution such as the Weibull or Pareto can be utilized to model the underlying continuous nature of the grouped response variable. The single- and two-factor model explained in this paper can easily be extended to a multifactor model to accommodate more than two explanatory variables.

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